Engineer Onboarding

Current Status

Verified on March 16, 2026 against the live Supabase-backed portal.

Smoke-tested engineer workflows:

- login and route protection
- engineer dashboard and system status surfaces
- settings access
- break-glass grant creation
- break-glass revoke
- redacted family view without active sensitive access

What Engineer Is For

Engineer is the platform-operations role. This role owns:

- system visibility
- admin governance
- incident tooling
- sensitive-access controls
- integration and release observability

Engineer is not a default "see everything" role. Billing and personal data stay redacted until a break-glass grant is opened for a specific support reason.

Core Boundaries

Engineer can:

- access every main section in the portal
- manage `admin`, `staff`, `ta`, and `instructor` accounts
- use role preview
- revoke sessions

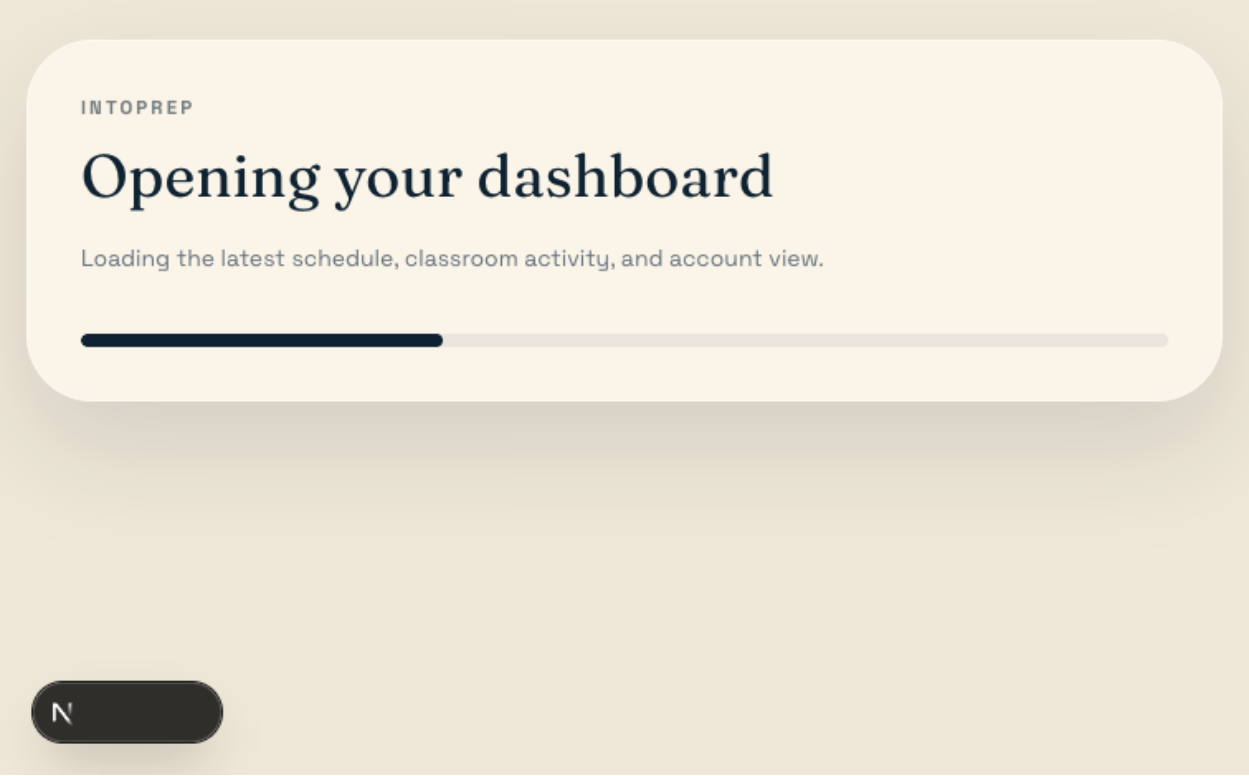
- review system health, drift, and integration status
- open and revoke break-glass access
- work incident, maintenance, and remediation tooling
- manage maintenance banners, change freeze, support notes, and feature flags
- inspect read-only schema and system state

Engineer cannot do by default:

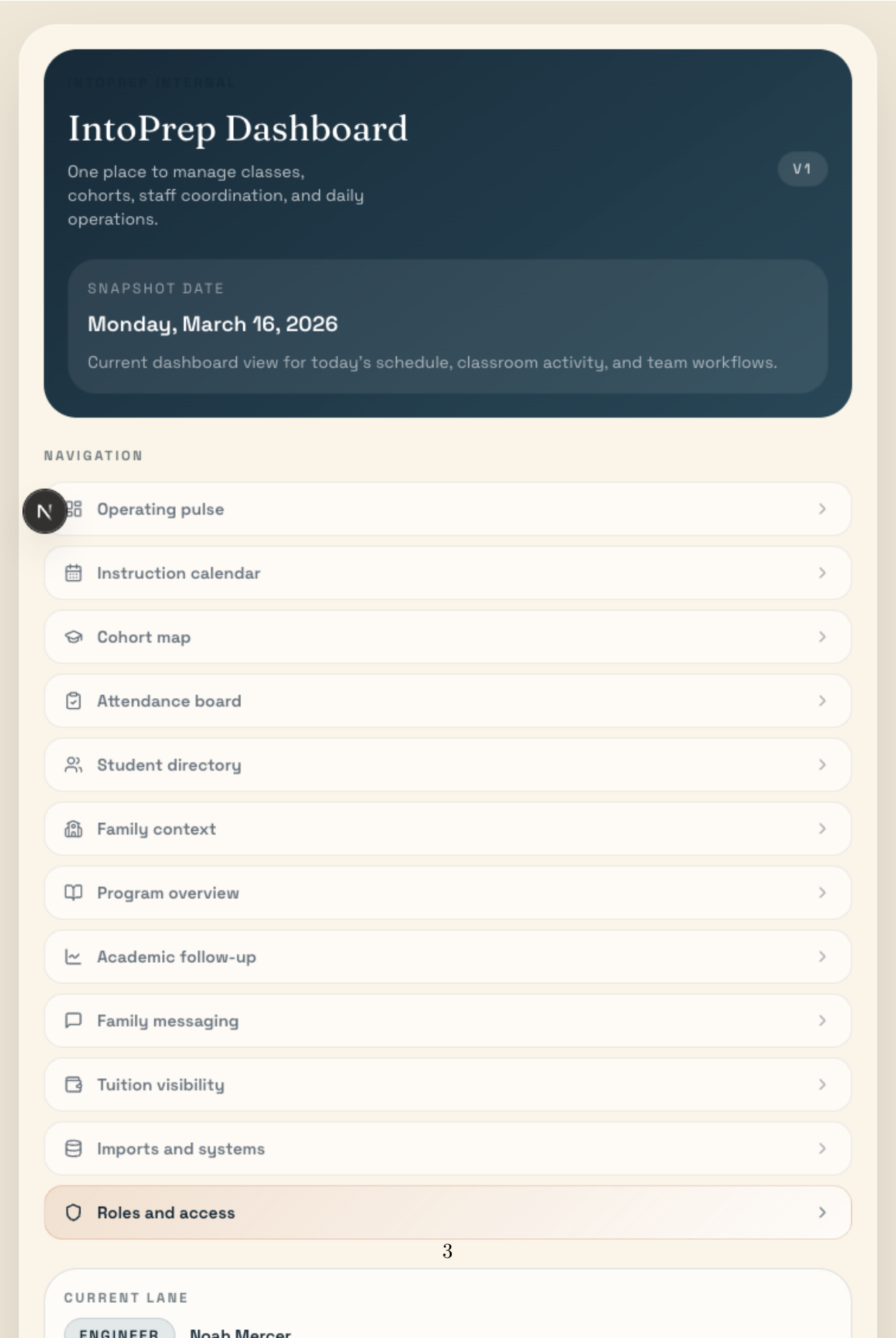
- view billing or family-sensitive detail without break-glass
- export sensitive data during break-glass
- manage engineer accounts in-app

Screens

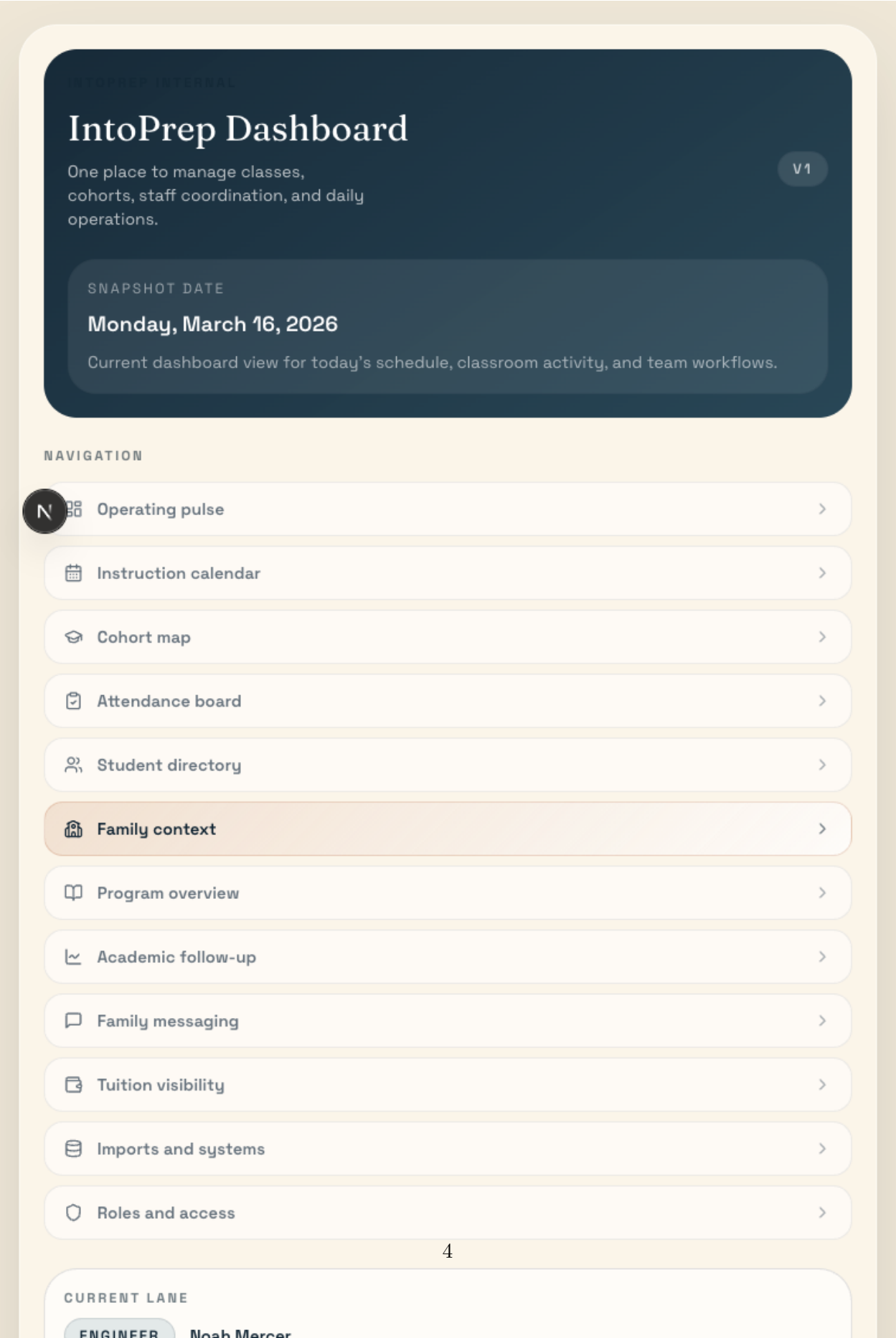
Engineer Dashboard



Engineer Settings



Engineer Family View With Redaction



First Login Walkthrough

1. Sign in at `/login`.
2. Confirm the left rail shows the full portal navigation.
3. Start on "Operating pulse" and review:
   - app version
   - Git commit
   - schema version
   - drift warnings
   - integration health
4. Open "Roles and access" and verify:
   - the managed user list loads
   - admin accounts are visible
   - engineer-only controls are present
5. Open "Family context" and confirm sensitive data is redacted until you intentionally open break-glass.

Daily Workflow

1. Check System Status

Use "Operating pulse" first.

Review:

- release version
- schema version
- cron health
- Google Forms source health
- QuickBooks source health
- email alert health
- recent sync warnings or failures

2. Review Governance

Open "Roles and access" and use this screen to:

- review active accounts
- provision or manage non-engineer roles
- manage admin accounts

- revoke sessions for compromised non-engineer accounts
- review audit activity

3. Use Break-Glass Only When Needed

Open a protected student, family, or billing context only when actively fixing a live issue.

Required inputs:

- a clear support reason
- the incident or support reference

Expected behavior:

- the grant is scoped
- a timer/banner appears
- the grant expires automatically
- the grant appears in the active break-glass monitor

4. Revoke Access When Done

From the active break-glass monitor, revoke the grant as soon as the support task is complete.

Complete Capability Map

Engineer is the only role that should see the full platform-control layer.

Use engineer for:

- platform health and release visibility
- schema and config drift review
- sync, import, and credential health review
- admin governance
- session revocation for compromised accounts
- break-glass access for sensitive family, student, or billing records
- incident acknowledgement, assignment, and maintenance-state actions
- remediation actions such as replay, retry, and repair
- maintenance banners and feature flags
- engineer-only support notes and schema inspection

Do not use engineer for:

- routine billing follow-up
- normal family communication

- day-to-day classroom work
- local production-data troubleshooting outside the hosted app

How To Use Break-Glass

1. Navigate to the protected family, student, or billing surface.
2. Select the break-glass action.
3. Enter:
    - a clear support reason
    - the incident or support reference
4. Confirm the grant opens.
5. Verify the banner/timer appears.
6. Complete the support task.
7. Revoke the grant from the monitor or let it expire.

How To Manage Admin Access

1. Open “Roles and access”.
2. Search for the target account.
3. Review current role, status, and recent audit context.
4. For destructive actions, read the confirmation text carefully.
5. Complete typed confirmation when prompted.
6. Confirm the audit log entry appears afterward.

Important:

- never remove the last active admin
- never use engineer actions casually for day-to-day school operations

How To Revoke A Session

1. Open “Roles and access”.
2. Find the affected user.
3. Confirm the user is not an engineer.
4. Choose the revoke-session action.
5. Confirm the action when prompted.
6. Check the audit log so there is a record of who forced the sign-out and why.

How To Use Role Preview

1. Start from the left rail role switcher or the preview links in the engineer lane.
2. Enter the target role preview such as admin, staff, TA, or instructor.
3. Watch for the preview banner.
4. Review the visible sections and workflow surfaces.
5. Exit preview before making live governance or support changes.

Preview is read-only and should be used to validate permissions and user experience, not to operate the business.

How To Work Incidents And Integrations

1. Start on "Operating pulse" and review warning and error states.
2. Open the relevant sync or integration surface.
3. Confirm source health, recent run history, and ownership.
4. If needed:
   - acknowledge the issue
   - assign the issue
   - add handoff context
   - retry or replay a failed step
   - pause a source or move it to maintenance mode
5. Add a support note if the incident needs shared context for later.

How To Use Maintenance Banner And Change Freeze

1. Use the maintenance banner when staff should see an internal portal-wide notice.
2. Keep banner wording plain, operational, and time-bound.
3. Use change freeze only when risky writes should be blocked during an incident.
4. Enter a clear reason and issue reference for any freeze.
5. Remove the banner or freeze as soon as the incident window closes.

How To Use Feature Flags, Support Notes, And Schema Inspector

- Feature flags: use them to roll out internal behavior changes by role without affecting everyone at once.
- Support notes: use them for engineer-only troubleshooting context tied to incidents, accounts, or integrations.

- Schema inspector: use it for read-only data and schema verification when the UI and database state appear out of sync.

These tools are for platform operations only and should not replace normal admin or staff workflow.

What To Watch During Incidents

- drift between app version and schema version
- incomplete integration setup
- failed or warning-state syncs
- repeated billing or intake mismatches
- unusual session or account activity in the audit log

Guardrails

- sensitive access is audited
- dangerous actions require explicit confirmation
- engineer activity should stay inside the hosted app for production investigations
- local development should stay on seeded or synthetic data

Verified Behavior

Verified in the live browser pass:

- engineer login succeeded
- engineer could access "Roles and access"
- engineer could create a break-glass grant for "family-bennett"
- engineer could revoke that same grant successfully
- sensitive family context stayed redacted before grant access

Not In Scope For Normal Engineer Use

- direct family operations
- routine billing follow-up
- staff task execution
- TA classroom support work
- instructor teaching workflows

Recommended First-Day Checklist

- review “Operating pulse”
- open “Roles and access”
- inspect the audit log
- test one break-glass grant and revoke flow in the demo data
- review current drift and sync warnings before governance changes